

Instructions

1. Follow the instructions in each question carefully.
2. A Jupyter notebook along with output for each cell is expected.
3. Any assignment submitted using other python IDEs are not considered for grading.
4. If there are any issues in accessing the links to datasets, you can search for the same dataset from any repositories and use them.
5. Incorrect Assignment Set submitted will not be considered.

NLP Assignment 2

Link to the Dataset:

https://drive.google.com/file/d/1d8eyQEtuxe_uPhV9ZROk-BWAZyziU3N84/view?usp=sharing

Description of Data:

The purpose of this analysis is to build a prediction model to predict whether a review on the restaurant is positive or negative. To do so, we will work on the Restaurant Review dataset. The dataset contains two columns, "Liked" and "Review". The liked can be 0 or 1.

1. Download the dataset and Create a dataframe named as **Rest_rev** then check the head, info, and describe methods on created dataframe **Rest_rev**. **(1 Marks)**
2. Perform pre-processing steps like Removing Punctuations, Numbers, and Special Characters, Stop Words in dataset. **(1 Marks)**
3. Normalize review by using Stemming or Lemmatization. **(1 Marks)**
4. Preprocessed **Review** should be included in the **Rest_rev** data frame as cleaned_Review. Plot word cloud for all classes of the Review. **(1 Marks)**
5. Create two objects **X** and **y**. **X** will be the 'cleaned_Review' column of **Rest_rev** data frame and **y** will be the 'liked' column. **(4 Marks)**
 - a. Create a Skip-gram object and split the data into training and testing sets. Train a Decision tree model. Display the confusion Matrix for both train and test.
 - b. Create a CBoW object and split the data into training and testing sets. Train a Decision tree model. Display the confusion Matrix for both train and test.
 - c. Compare Skip-gram and CBow. **Answer without justification will not be awarded marks.**
6. Display the HMM POS tagging on the first 4 rows of 'cleaned_Review'. **(2 Marks)**